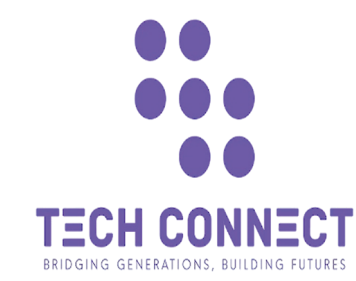




Deep Learning-Based Depression Detection from Brain MRI

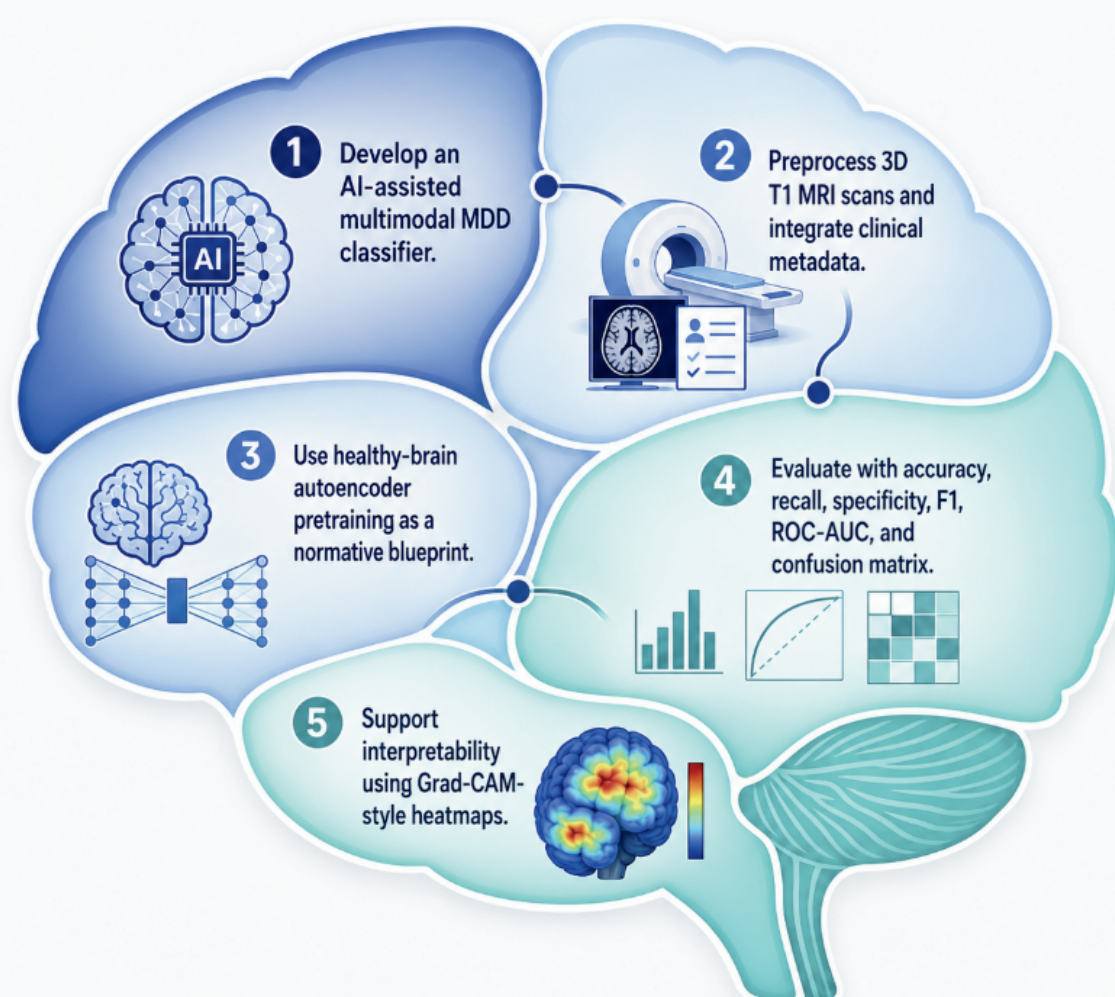


Team Members FAHAD AL NAHI ,IBRAHIM HURUBI ,FAHAD AL-MUBARAK ,ABDULLAH ALSHAHARANI ,ALI MUYINI **Supervisor** Dr. Mohamed Ghouse

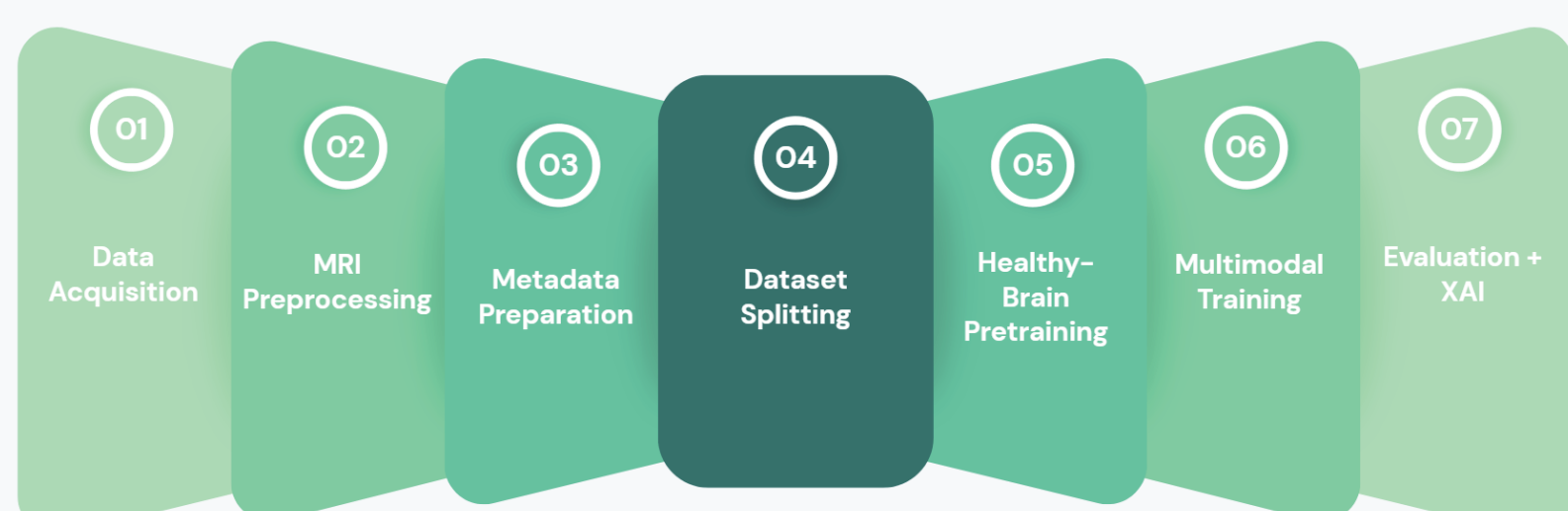
ABSTRACT

Major Depressive Disorder (MDD) is one of the most common mental health disorders and is usually diagnosed through clinical interviews and self-report questionnaires. However, these methods can be subjective and may lead to delayed or inconsistent diagnosis. This project presents an AI-assisted framework for detecting MDD using T1-weighted 3D brain MRI scans and clinical metadata. The proposed system uses a multimodal deep learning model that combines MRI-based anatomical features with patient-level clinical information such as age, sex, education, and HAMD score. The model applies a 3D CNN with Squeeze-and-Excitation attention, a clinical MLP branch, and a late-fusion classifier to distinguish between Healthy Control and MDD subjects. The system also supports Grad-CAM-style visualization to highlight important brain regions involved in the prediction, making the model more interpretable as a research prototype.

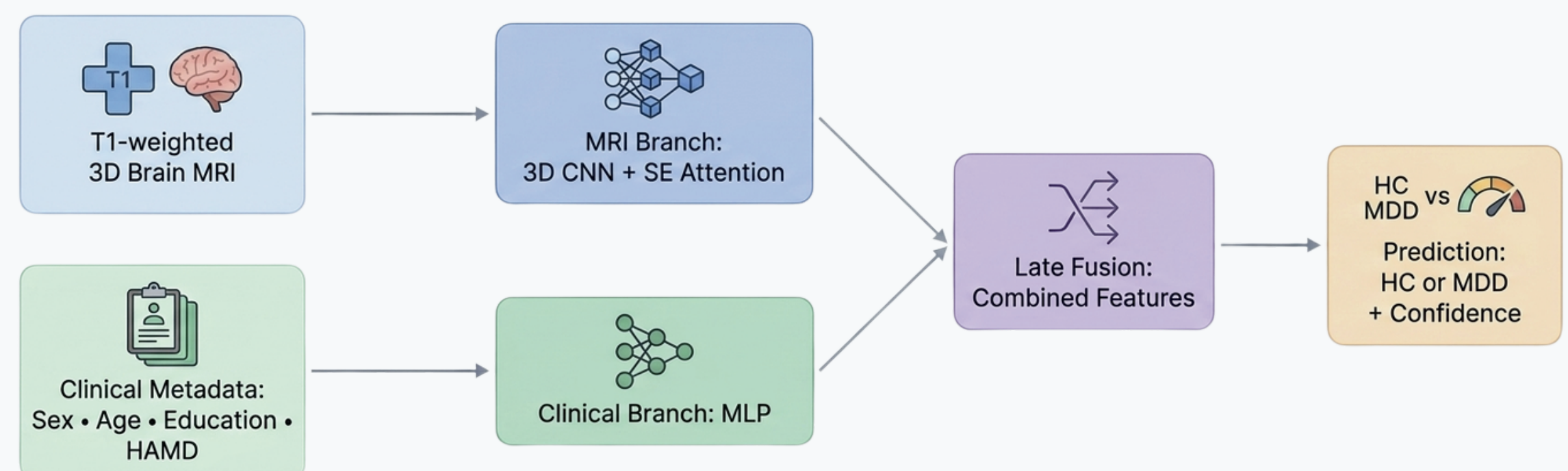
OBJECTIVES



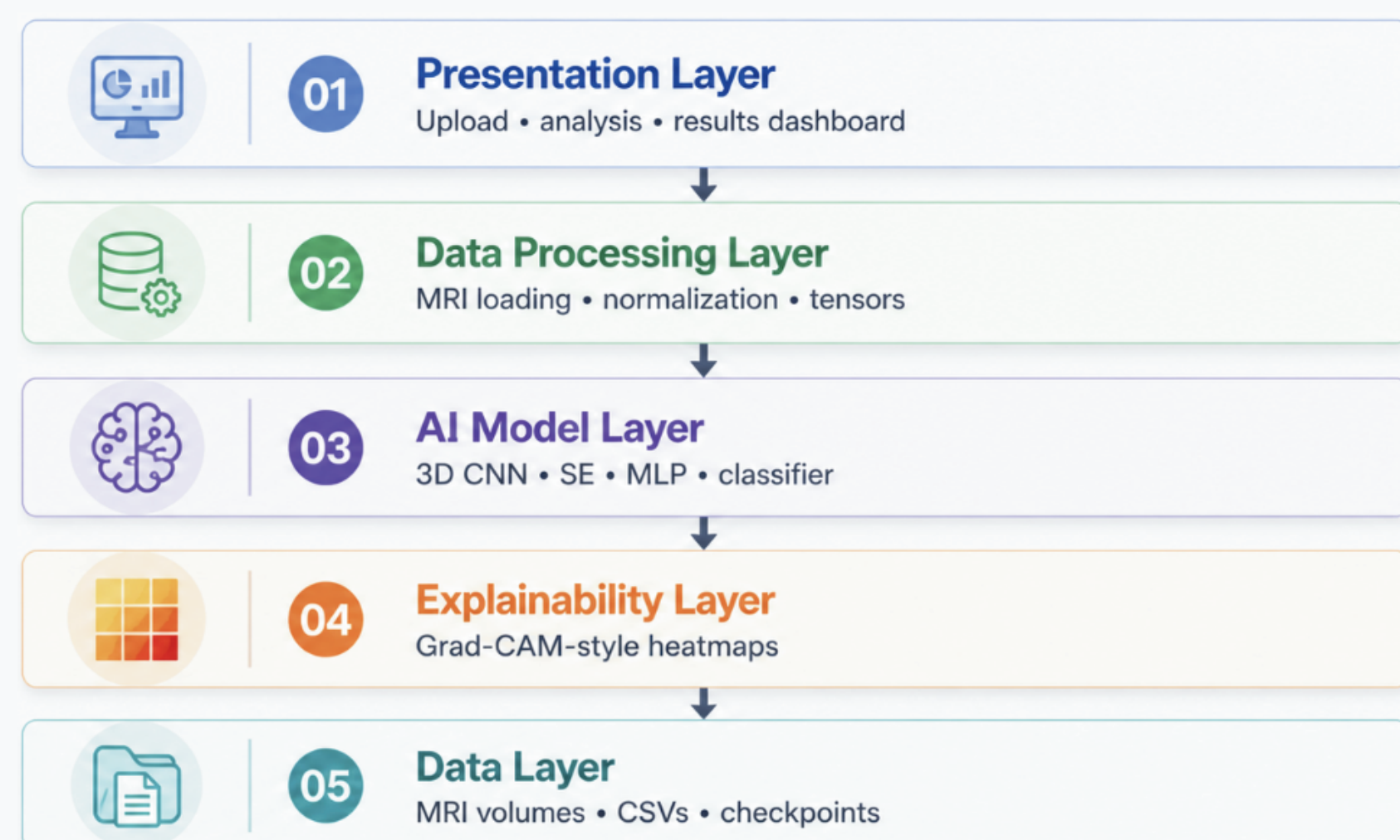
METHODOLOGY



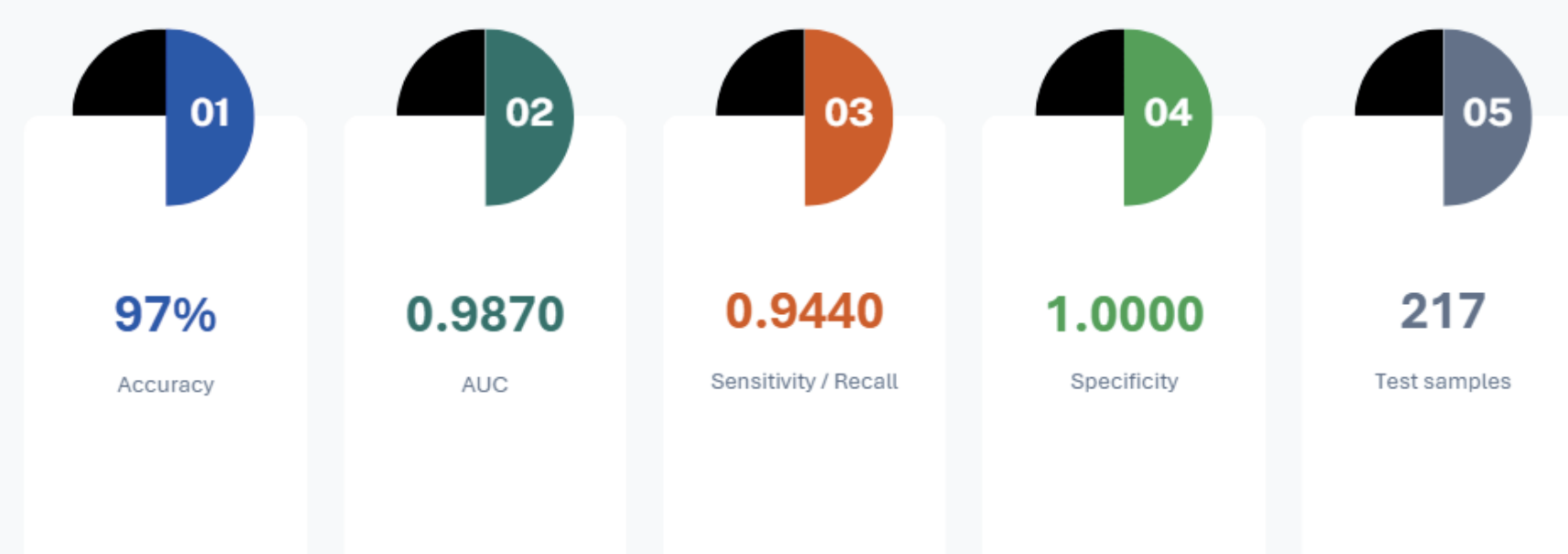
MODEL ARCHITECTURE



SYSTEM ARCHITECTURE



RESULTS



TOOLS & TECHNOLOGIES



CONCLUSION

This project shows that deep learning can support MDD detection using 3D brain MRI and clinical metadata. It provides a promising research prototype, but requires further validation before clinical use.